Amazon EFS — 30 Important Interview Questions & Answers

1. What is Amazon EFS?
Answer:
Amazon EFS stands for **Amazon Elastic File System**. It is a managed file storage service in AWS that can be shared by multiple EC2 instances at the same time. EFS automatically grows and shrinks based on the amount of data stored. It is mainly used when multiple servers need to access the same files.
For example:
Suppose we have three EC2 servers running the same application. If all three servers need to access the same uploaded files, we can store those files in EFS instead of keeping separate copies on each EC2 server.
Example:

```
        EFS
      /  |  \
     ↓   ↓   ↓
   EC2-1  EC2-2  EC2-3
```

Simple ga gurthupettuko:
EFS = Shared File Storage for multiple EC2 instances

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2. Why do we use EFS?
Answer:
We use EFS when multiple servers need to access the same files. Unlike local storage, EFS can be mounted by multiple EC2 instances. It also manages storage capacity automatically, so we do not need to manually increase the file system size.
For example:
Suppose an application has multiple EC2 servers and users upload profile pictures. All EC2 servers should be able to access those pictures, so we can store them in EFS.
Example:

```
Users
 ↓
Application Servers
 ↓
EFS
 ↓
Shared Files
```

Simple ga gurthupettuko:
EFS = One shared file system for many servers

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3. Is EFS block storage or file storage?
Answer:
EFS is a **file storage service**. It stores data as files and directories, similar to a normal file system. EC2 instances can mount the EFS file system and access files using normal file-system operations.
For example:
If we have folders like /uploads, /images, and /documents, EFS can store and share these folders between multiple EC2 instances.
Example:

```
EFS
 ├── images
 ├── uploads
 └── documents
```

Simple ga gurthupettuko:
EFS = File storage

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4. Can multiple EC2 instances access the same EFS?
Answer:
Yes. Multiple EC2 instances can mount the same EFS file system at the same time. This is one of the main reasons we use EFS. The instances can read and write files from the shared file system.
For example:
If EC2-1 uploads a file to EFS, EC2-2 can access that same file through the EFS mount.
Example:

```
      EFS
    /  |  \
   ↓   ↓   ↓
 EC2-1 EC2-2 EC2-3
```

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5. How does EC2 connect to EFS?
Answer:
EC2 connects to EFS by mounting the EFS file system. The EC2 instance needs network connectivity to EFS and the required security group rules. After mounting, EFS appears like a directory on the EC2 server.
For example:
We can mount EFS on an EC2 server at a location such as /mnt/efs, and the application can use that directory to store files.
Example:
EC2

↓
Mount EFS
↓
/mnt/efs
↓
Files

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6. What is an EFS File System?
Answer:
An EFS File System is the main storage system where our files and directories are stored. When we create EFS, the file system becomes the shared storage that EC2 instances can mount and use.
For example:
If we create an EFS file system called application-files, all required EC2 instances can connect to that file system.
Example:
EFS File System
↓
Application Files
↓
EC2 Instances

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7. What is an EFS Mount Target?
Answer:
A Mount Target provides a network endpoint through which EC2 instances access an EFS file system. Mount targets are created inside VPC subnets. For high availability, we normally create mount targets in the Availability Zones where our application servers run.
For example:
If EC2 servers are running in multiple Availability Zones, we can create EFS mount targets in those Availability Zones.
Example:
VPC
|
EFS
|

| |
Mount Target Mount Target
AZ-1 AZ-2

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8. Why do we need Mount Targets?
Answer:
EC2 instances need a network path to access EFS. Mount targets provide that network path inside the VPC. Without the required mount target and network access, the EC2 instance cannot properly mount the EFS file system.
For example:
EC2 in AZ-1 can connect to the EFS mount target available in the required subnet and access the shared file system.

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9. What is an EFS Access Point?
Answer:
An EFS Access Point provides an application-specific way to access an EFS file system. It can define a root directory and permissions for that application. This is useful when multiple applications share the same EFS file system.
For example:
One application can use /app1 while another application uses /app2 through different access points.
Example:
EFS
/ \
Access Access
Point-1 Point-2
↓ ↓
App-1 App-2

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10. Why do we use EFS Access Points?
Answer:
Access Points help control how applications access specific parts of an EFS file system. They can simplify permissions and provide a separate starting directory for an application. They are especially useful when multiple applications share one EFS file system.
For example:
A company can use one EFS file system but give different applications their own directories through separate access points.

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11. What are EFS Storage Classes?
Answer:
EFS provides different storage classes based on how frequently data is accessed. The main idea is to keep frequently used files in a faster access class and less frequently used files in a lower-cost class. EFS can also move eligible files between classes using lifecycle management.

For example:
Frequently used application files can stay in the main storage class, while old files can move to an infrequent-access class.

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12. What is EFS Lifecycle Management?
Answer:
Lifecycle Management automatically moves files between EFS storage classes based on how long they have not been accessed. This helps reduce storage cost. We can configure the lifecycle policy according to the application's requirements.
For example:
If a file has not been accessed for a certain number of days, EFS can automatically move it to a lower-cost storage class.
Example:
Frequently Used File
↓
Standard Storage

Not Used for Long Time
↓
Infrequent Access Storage

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13. What is EFS Throughput?
Answer:
Throughput is the amount of data that the EFS file system can read or write over a period of time. EFS provides different throughput modes depending on the workload. We choose the appropriate mode based on application requirements.
For example:
If an application regularly transfers large amounts of files, we need to choose a throughput configuration that can handle that workload.

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14. What is EFS Performance Mode?
Answer:
Performance Mode controls how the EFS file system is optimized for performance. EFS provides modes designed for different application requirements. The correct choice depends on the type of workload and expected file operations.
For example:
A normal application may use the default performance configuration, while a highly demanding application may require a different configuration based on its workload.

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15. What is EFS Elastic Throughput?
Answer:
Elastic Throughput allows EFS throughput to automatically increase or decrease based on the workload. This is useful when application traffic changes frequently. We do not need to manually predict the exact throughput requirement all the time.
For example:
During normal hours the application may use less throughput, but during a traffic spike EFS can provide more throughput based on the workload.

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16. What is EFS Replication?
Answer:
EFS Replication allows data from one EFS file system to be replicated to another EFS file system. It can be useful for disaster recovery and keeping a copy of data in another AWS Region.
For example:
We can keep the main EFS file system in one Region and replicate the data to another Region for disaster recovery.
Example:
EFS - Region A
↓
Replication
↓
EFS - Region B

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17. Is EFS Regional or tied to one Availability Zone?
Answer:
Standard EFS is designed as a Regional service and can be accessed across Availability Zones in a Region. This makes it useful for applications running on multiple EC2 instances across different AZs.
For example:
EC2 instances in AZ-1 and AZ-2 can access the same EFS file system.

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18. How do you secure EFS?
Answer:
We can secure EFS using VPC networking, security groups, IAM-based controls where supported, EFS access points, and encryption. We should allow only the required application servers to access the file system.
For example:
EC2 Security Group
↓
EFS Security Group
↓

EFS

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19. What port does EFS use?
Answer:
EFS normally uses **NFS**, and NFS uses TCP port **2049**. The security group associated with EFS must allow the required NFS traffic from the application servers.
For example:
If EC2 needs to mount EFS, the EFS security group should allow TCP 2049 from the appropriate EC2 security group.

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20. What is encryption in EFS?
Answer:
EFS supports encryption for data stored in the file system. It can also support encryption while data is transferred between the EC2 instance and EFS. Encryption helps protect sensitive application files.
For example:
EC2
↓
Encrypted Connection
↓
EFS
↓
Encrypted Data

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21. What is the difference between EFS and EBS?
Answer:
EBS is block storage mainly attached to EC2 instances, while EFS is shared file storage that can be accessed by multiple EC2 instances. EBS is commonly used for operating system disks and application data, while EFS is useful when multiple servers need the same files.
For example:
EBS → One EC2 / Block Storage

EFS → Multiple EC2 / Shared File Storage

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22. What is the difference between EFS and S3?
Answer:
EFS is a file system that applications can mount and use like a normal file system. S3 is object storage where data is stored as objects inside buckets. EFS is useful for applications that need file-system access, while S3 is useful for storing objects such as images, videos, backups, and documents.
For example:
EFS → Shared application file system

S3 → Object storage

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23. Can EFS automatically scale?
Answer:
Yes. EFS automatically grows as we add more files and reduces storage usage when data is removed. We do not normally need to manually select a fixed storage size like we do with many block-storage setups.
For example:
If our application grows from 100 GB to 500 GB of files, EFS can automatically handle the increased storage requirement.

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24. What is EFS Backup?
Answer:
EFS can be protected using AWS Backup. AWS Backup can help create and manage backups according to configured backup policies. This is useful for protecting important file-system data.
For example:
EFS
↓
AWS Backup
↓
Backup
↓
Recovery when required

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25. What is EFS Monitoring?
Answer:
EFS can be monitored using Amazon CloudWatch and other AWS monitoring features. We can monitor file-system activity and performance-related metrics. Monitoring helps us identify unusual usage or performance problems.
For example:
EFS
↓
CloudWatch
↓

Metrics
↓
Monitoring / Alerts

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26. What is EFS File System Policy?
Answer:
An EFS File System Policy is a resource-based policy attached to the EFS file system. It can control which AWS identities or access methods are allowed to perform supported actions on the file system. It adds another security layer along with network controls.
For example:
We can create a policy that allows only approved application access to the EFS file system.

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27. What is the EFS Console?
Answer:
The EFS console is used to create, configure, monitor, and manage EFS file systems. It contains sections for file systems, access points, replication, backup-related options, and other configuration areas.
Example:
EFS Console
│
├── File Systems
├── Access Points
├── Replication
├── Backup
└── Monitoring

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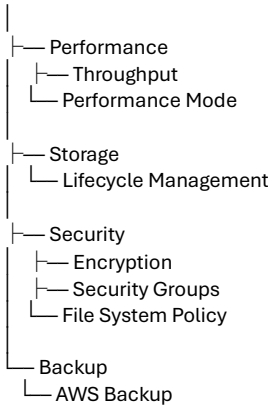
28. What do we see inside an EFS File System?
Answer:
When we open an EFS file system, we can see information such as file-system ID, storage usage, performance settings, throughput settings, mount targets, network configuration, lifecycle settings, encryption, and access points.
For example:
EFS File System
↓
Network
↓
Performance
↓
Storage
↓
Security
↓
Access Points

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29. What is the most important thing to remember about EFS?
Answer:
EFS is mainly used when multiple EC2 instances need shared file storage. It automatically manages storage capacity and can be used across Availability Zones. EFS also provides security, lifecycle management, access points, and backup options.
For example:
Suppose we have three EC2 servers running the same web application. Instead of storing uploaded files separately on every server, we can store them in EFS so all three servers can access the same files.
Example:
EFS
/ | \
↓ ↓ ↓
EC2-1 EC2-2 EC2-3

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30. EFS Console — Easy Memory Map
Answer:
Remember the EFS console like this:
EFS
│
├── File Systems
│ └── Main EFS Storage
│
├── Access Points
│ └── Application Access
│
├── Replication
│ └── Copy Data
│
├── Network
│ └── Mount Targets



Simple ga gurthupettuko:
EFS = Shared Files + Mount Targets + Access Points + Lifecycle + Performance + Security + Backup